



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
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Engineering, Built Environment and IT
Department of Computer Science
Imperative Programming
COS 132
Semester test 1

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Examiners

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Instructions

1. Read the question paper carefully and answer all the questions.
2. Write all your answers in the answer books provided. All answers must be written indelible pen.
3. The assessment opportunity comprises of 6 questions on 6 pages. Refer to the table below for the marks per question.
4. This is a closed book assessment.
5. One and a half (1.5) hours have been allocated for you to complete this paper.
6. In the event of loadshedding, extra time will be given depending on how long it takes the generators to kick in.

Integrity statement:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

Question:	1	2	3	4	5	6	Total
Marks:	18	8	7	11	8	15	67

Full marks is: 60

1. (a) For each of the statements below, state whether the statement is True or False.
 - i. The semantic rules of a language tell you what is legal and what is not legal. (1)
 - ii. In C++, the relational operator `>` returns true if the left operand is greater than the right operand. (1)
 - iii. If you refer to an identifier without declaring it, the compiler will generate an error message. (1)
 - iv. The `cin` operator should be used, if you would like to read a string containing blanks (1)
 - v. An expression such as `pow(2,3)` is called an operator call. (1)
 - vi. To use the function `sqrt`, the `cmath` library must be included. (1)
 - vii. When comparing two strings, the relational operator `>`, will return true if the lengths of the strings are the same. (1)
- (b) For each of the questions below, choose the most correct option(s).
 - i. Which one of the following Linux bash commands creates a new, empty file or updates the timestamp of an existing file? (1)
 - A. `newtime`
 - B. `nano`
 - C. `touch`
 - D. `cp`
 - ii. Which of the following statements about functions in C++ is true? (1)
 - A. Functions cannot return values.
 - B. Functions must be defined before they are used.
 - C. Functions cannot have parameters.
 - D. Functions cannot call other functions.
 - iii. Which one of the following is true regarding function overloading in C++? (1)
 - A. Functions with the same name must have the same return type.
 - B. Functions with the same name must have different parameter types.
 - C. Functions with the same name must have the same number of parameters.
 - D. Functions with the same name must have the same implementation.
 - iv. Which one of the following C++ code statements is a valid example of using the `static_cast` operator? (1)
 - A. `static_cast<int>("123")`
 - B. `static_cast<float>(5 / 2)`
 - C. `static_cast<char>(65)`
 - D. `static_cast<double>("7.8")`
 - v. Which of the following C++ code statements will result in a compiler error? (1)
 - A. `return a == 0;`
 - B. `return 0 == a;`
 - C. `return a = 0;`
 - D. `return 0 = a;`
 - vi. Which relational operator(s) is/are used to check if the right operand is less than or equal to left operand? (1)
 - A. `<`
 - B. `<=`
 - C. `>`
 - D. `>=`
- (c) For each of the following statements, fill in the missing words.
 - i. A(n) _____ operator has only one operand. (1)
 - ii. You can use _____ to override the order of precedence rules. (1)
 - iii. The header file _____ needs to be included to use the `setw` function. (1)
 - iv. The maximum number of significant digits in a number is called the _____. (1)
 - v. A variable is said to be _____, the first time a value is placed in it. (1)
2. For each of the following programs, study the program and provide the output for the program.
 - (a)

```
#include <iostream>
using namespace std;
int main() {
    double solution = 10 + 2 * 3 - 4 + 6 / 2;
    cout << solution;
    return 0;
}
```

 (2)

(b) `#include <iostream>`
`using namespace std;`

(2)

```
int main() {
    int m = 3;
    int n;
    n = m++ * 4;
    cout << n;
    return 0;
}
```

(c) `#include <iostream>`
`using namespace std;`

(2)

```
int myfunc(int x, int y) {
    return (x % y);
}

double myfunc(double p, double q) {
    return (p / q);
}

int main() {
    int a = 9, b = 4;
    double c = 8.0, d = 2.5;

    cout << myfunc(a, b) << " ";
    cout << myfunc(c, d);

    return 0;
}
```

(d) `#include <iostream>`
`using namespace std;`

(2)

```
bool myAND(bool, bool = true);
bool myOR(bool, bool = true);

int main() {
    int v1 = 1;
    int v2 = 0;
    cout << myAND(myOR(v1,v2),v2) << endl;
    cout << myAND(myOR(v1,v2)) << endl;
    return 0;
}

bool myAND(bool a, bool b) {
    return a && b;
}

bool myOR(bool a, bool b) {
    return a || b;
}
```

3. Consider the following C++ program.

(7)

```

1 #include <iostream>
2 using namespace std;
3 int main() {
4     cout << " This line has a syntax error. " << endl;
5     cout << " This line also has a syntax error. << endl;
6     int x = 10;
7     double y = 20.5;
8     int z = x + y;
9     double w = y / (10 - x);
10    cout<<w >>" \n ";
11    return 0;
12 }

```

Identify the three syntax and two semantic errors in the given code. For each of the errors, provide the error type (Syntax or Semantic), the line number on which the error occurs and a short description or explanation of the error.

4. (a) You are hosting a dinner party and want to plan a multi-course meal that flows smoothly from appetisers to desserts. Apply the algorithmic thinking process to write an algorithm to organise the courses and ensure a balanced and satisfying dining experience for your guests. (6)

- (b) Given the following C++ code: (5)

```

#include <iostream>
using namespace std;
int main() {
    int number;
    cout << "Enter a number: ";
    cin >> number;
    if (number % 2 == 0) {
        cout << "Even Number." << endl;
    } else {
        cout << "Odd Number." << endl;
    }
    return 0;
}

```

Draw a flowchart that captures the algorithm represented by the given code.

5. Consider the following scenario in a geometry calculation program:

Suppose we have the following variables representing the radius and height of a cylinder:

```

int radius; // in meters
int height; // in meters
double volume; // in cubic meters

```

You are given the formula to calculate the volume of a cylinder:

$$V = \pi r^2 h$$

Answer the following questions based on this scenario.

- Write the compiler preprocessor directive should be included to enable input and output operations? (1)
 - What will the most appropriate method be to define π (PI)? (1)
 - Based on your answer above, write a single C++ statement to define PI. (2)
 - Write a single C++ statement to read the radius and height of the cylinder from standard input. (2)
 - Write a single C++ statement to calculate the volume of the cylinder. Use the provided formula to calculate the volume and use an appropriate type cast expression to ensure that the volume is calculated correctly. (2)
6. Consider the following header (.h) and implementation files (.cpp) and answer the question that follow.

Header file: Advanced_Maths.h

```

#ifndef Advanced_Maths_h
#define Advanced_Maths_h

```

```

float squareRoot(float);
float areaOfCircle(float);
float areaOfTriangle(float, float);

```

```
#endif
```

Implementation file: Advanced_Maths.cpp

```

#include "Advanced_Maths.h"
#include "Constants.h"
#include <math.h>

float squareRoot(float x){
    //Implementation here
}

float areaOfCircle(float radius){
    //Implementation here
}

float areaOfTriangle(float base, float height){
    //Implementation here
}

```

Header file: Maths.h

```

#ifndef Maths_h
#define Maths_h

    float sum(float, float);
    float subtract(float, float);
    float multiply(float, float);
    float divide(float, float);

#endif

```

Header file: Maths.cpp

```

#include "Maths.h"

float sum(float x, float y){
    //Implementation here
}

float subtract(float x, float y){
    //Implementation here
}

float multiply(float x, float y){
    //Implementation here
}

float divide(float x, float y){
    //Implementation here
}

```

You may assume that a constant PI has been defined in a header file, Constants.h. The main function is defined in a file called Calculator.cpp.

Main function Calculator.cpp

```

#include <iostream>

#include "Advanced_Maths.h"
#include "Maths.h"

```

```
using namespace std;
```

```
int main(){  
    cout<<sum(2.7,3.2)<<endl;  
    cout<<subtract(3.3,1.9)<<endl;  
    cout<<multiply(4.7,7.8)<<endl;  
    cout<<divide(23.5,12.7)<<endl;  
    cout<<squareRoot(123.4)<<endl;  
    cout<<areaOfCircle(12.67)<<endl;  
    cout<<areaOfTriangle(sum(123.4,5),multiply(23.2,21))<<endl;  
    return 0;  
}
```

- (a) Provide the implementations for each of the following functions. Your implementation may only use functions defined in Maths.h and Advanced_Maths.h.
- i. The area of a triangle (areaOfTriangle). The formula for the area of a triangle is given by: $AreaTriangle = \frac{1}{2} \times base \times \perp height$. (3)
 - ii. The area of a circle (areaOfCircle). The formula is given by: $AreaCircle = \pi \times radius^2$ (3)
- (b)
- i. Write the makefile entry needed to compile Advanced_Maths.o and Maths.o. Indicate in your answer where a mandatory tab(s) are to be included. (5)
 - ii. Assume that the Calculator.o, Advanced_Maths.o and Maths.o exist. Write the makefile entry needed to compile the program into an executable called Main. (2)
 - iii. Which program is used to run the makefile? (1)
 - iv. Provide the command to run the executable. (1)